RASHMI DANGOL

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PROFESSIONAL SUMMARY

Innovative geospatial enthusiast and FAA-certified Remote Pilot with experience in environmental and spatial data collection, geospatial technologies and remote sensing applications. Skilled in field-based research, GIS, spatial data analysis, and scientific reporting. Proven track record at **Syngenta** and **Bayer** to solve real world agricultural challenges and optimize natural resources management.

EDUCATION

Graduate Certificate in Geographic Information Science (GIS) | Texas A&M University, TX — Dec 2025

Master of Science in Applied Science and Technology | Illinois State University, Normal, IL — May 2024

Bachelor of Science in Agricultural Systems | Agriculture and Forestry University, Nepal — Nov 2019

EXPERIENCE

Research Assistant (Geospatial focus) | Texas A&M University — June 2024-Present

- Utilize GIS software, drones, satellite imagery and sensors for identification of problematic invasive plant species supporting vegetation monitoring and effective rangeland management.
- Build interactive web-based data visualization applications using ArcGIS Dashboards, Experience Builder, Story Maps to visualize environmental and crop data.
- Perform geospatial analysis for drone images and land cover data including orthomosaic, vegetative index calculations, and buffer analysis in ArcGIS Pro and ArcGIS Notebooks.

Product Placement Phenotyping Intern | Syngenta, Malta, IL — May 2025-Aug 2025

- Created shapefiles and customized field maps in ArcGIS Pro and Google Earth Pro to streamline project planning.
- Led DJI Matrice 300 RTK drone flight operations-captured and analyzed high-resolution aerial images to deliver new index for vegetation health ratings.
- Collaborated on a Python-based (ArcPy, Geopandas) machine-learning image processing pipeline applying QA/QC protocols to raw aerial imagery to build accurate predictive models for field decision support.

Graduate Research Assistant | Illinois State University — May 2022-May 2024

- Conducted field-based soil investigations involving soil sampling and data collection to evaluate soil pH, soil organic matter variability across different cropping systems
- Prepared project presentations, report writing and data summaries communicating scientific findings to technical and non-technical audiences and international conferences.

Seed Technology Intern | Bayer, Oxnard, CA — May 2023-Aug 2023

- Managed soil analysis and tissue culture data using the Smart Farm Plus application, conducted seed priming lab experiments to support seed quality protocols.
- Executed field data collection for yield performance using GPS rovers, ensuring high data integrity for the technical product sheets.

Digital Enumerator/Surveyor | International Maize and Wheat Improvement Center (**CIMMYT**) — May2021-Sep 2021

- Led a survey team of six members for the National Seed and Fertilizer Project (NSAF), leveraging Kobo Toolbox to digitize data collection and evaluate the effectiveness of ICT tools among rural farmers.

AWARDS AND CERTIFICATIONS

- **FAA Part 107 Certified Remote Pilot** for UAV Operations (Commercial Drone License) — Sep 2024
- **James L. Fisher 2024 Outstanding Thesis Award-** Illinois State University — Nov 2024
- **Outstanding Graduate Student Recognition-** Illinois State University — Apr 2024

SKILLS AND TRAINING

Proficient in Microsoft Office, ArcGIS Pro, Pix4D Mapper, QGIS, Python, SPSS, SAS, Computer Vision (YOLO), ArcMap, Power BI, ArcGIS Dashboards, StoryMaps, ArcGIS Experience Builder

Field Operations and Tools: Drones, GPS Rover, High-Throughput Phenotyping, Field Mapping, Emlid Flow